

KEETMANSHOOP, Namibia

WMO: 683120

Lat: 26.540S

Lon: 18.111E

Elev: 1069

StdP: 89.13

Time Zone: 2.00 (E02)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.1	4.8	-15.0	1.2	18.4	-12.9	1.4	18.9	10.6	19.7	9.5	19.2	2.1	70	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	38.6	17.4	37.3	17.0	36.2	16.8	21.0	30.0	20.2	29.2	19.6	28.7	3.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.6	23.0	17.9	14.6	23.7	16.8	13.6	24.3	66.1	29.0	63.3	29.3	60.9	28.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	28.0	27.6	25.8	22.1	19.0	15.0	14.8	16.2	20.1	23.0	24.8	27.0
	DBStd	5.91	2.66	2.54	3.14	3.56	3.93	3.46	3.71	4.00	4.87	4.21	3.56	2.82
	HDD10.0	16	0	0	0	0	0	5	7	3	0	0	0	0
	HDD18.3	419	0	0	2	11	41	106	118	88	39	12	2	0
	CDD10.0	4373	559	492	491	364	279	155	155	197	304	404	444	528
	CDD18.3	1736	301	258	234	125	62	6	7	24	93	159	196	270
	CDH23.3	21408	4012	3174	2668	1202	602	102	126	321	1096	1987	2537	3581
CDH26.7	11189	2387	1738	1344	478	183	8	18	76	501	1011	1370	2075	
Wind	WSAvg	3.2	3.5	2.6	2.6	2.6	2.7	3.4	3.3	3.5	3.3	3.6	3.7	3.7
Precipitation	PrecAvg	150	27	39	33	14	5	2	1	0	3	6	8	10
	PrecMax	482	221	144	140	60	39	15	21	6	49	41	47	38
	PrecMin	29	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	92	40	39	30	15	10	3	4	1	9	11	11	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	38.9	37.4	35.2	32.7	27.8	28.8	30.7	36.2	38.0	39.5	39.7
		MCWB	17.9	16.9	16.8	16.1	14.5	12.0	11.6	12.8	14.9	16.2	17.0	17.6
	2%	DB	38.3	37.1	35.6	33.2	30.1	25.9	26.1	28.7	33.9	36.0	37.1	38.1
		MCWB	17.7	17.1	17.2	15.9	13.7	11.3	10.7	11.7	14.6	15.7	16.8	17.1
	5%	DB	37.1	35.8	34.2	31.3	28.5	24.4	24.3	26.9	31.8	34.2	35.7	36.8
		MCWB	17.2	17.1	17.1	15.6	13.3	11.0	10.3	11.0	13.8	15.3	16.6	16.9
	10%	DB	35.8	34.5	32.9	29.6	26.6	22.8	22.6	24.9	29.7	32.2	33.9	35.2
		MCWB	17.1	17.2	16.8	15.3	12.8	10.4	9.6	10.4	13.2	14.8	15.3	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	21.8	21.4	19.3	15.7	14.6	12.6	13.4	16.7	19.7	22.1	20.8
		MCDB	29.6	29.7	28.2	26.8	27.2	22.5	25.7	27.9	32.5	32.4	34.5	31.1
	2%	WB	20.2	20.9	20.3	18.3	14.7	12.6	11.5	12.3	15.3	18.0	19.9	19.7
		MCDB	29.2	29.4	27.8	24.9	27.0	22.7	24.1	26.2	30.6	29.9	30.9	30.2
	5%	WB	19.4	20.3	19.6	17.6	14.0	11.7	10.7	11.5	14.4	16.7	18.0	18.9
		MCDB	30.0	29.0	27.0	25.0	26.5	22.3	23.1	25.1	30.0	30.5	30.0	30.0
	10%	WB	18.7	19.6	19.0	16.9	13.2	10.9	10.0	10.7	13.4	15.7	16.7	18.1
		MCDB	30.8	28.5	26.6	25.5	25.2	21.5	21.7	23.7	28.3	29.4	30.5	31.1
Mean Daily Temperature Range	5% DB	MDBR	15.1	13.5	13.0	12.9	13.4	13.8	14.2	15.0	15.0	15.7	16.2	15.8
		MCDBR	16.1	15.0	14.2	14.8	14.6	15.0	15.9	16.3	16.5	17.5	17.7	16.7
		MCWBR	5.3	5.2	4.9	5.8	6.0	6.7	7.4	7.4	6.9	7.3	8.2	5.4
	5% WB	MCDBR	13.9	12.2	12.2	11.4	13.8	14.2	14.7	15.6	15.7	16.1	15.7	14.5
MCWBR		4.8	4.6	4.0	4.3	5.8	6.9	6.9	7.1	7.0	7.7	8.7	5.6	
Clear-Sky Solar Irradiance	taub	0.333	0.344	0.328	0.302	0.272	0.257	0.260	0.290	0.327	0.340	0.305	0.312	
	taud	2.520	2.500	2.521	2.571	2.623	2.626	2.605	2.470	2.345	2.343	2.528	2.540	
	Ebn at Noon	1010	982	969	951	942	938	947	949	953	974	1031	1032	
	Edh at Noon	113	112	104	90	77	73	78	98	122	130	111	111	
All-Sky Solar Radiation	RadAvg	8.26	7.50	6.60	5.48	4.57	4.12	4.38	5.28	6.42	7.49	8.43	8.72	
	RadStd	0.35	0.45	0.24	0.20	0.12	0.11	0.09	0.17	0.20	0.23	0.25	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page